From Seizure Onset Zone Localization to Outcome: Hybrid GNN–Random Forest Models for Epilepsy Surgery Planning

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Abstract

Accurate localization of the seizure onset zone (SOZ) is arguably the most consequential step in planning epilepsy surgery. Intracranial EEG provides the spatial and temporal resolution needed for this task, yet manual review remains time-consuming and prone to inter-rater disagreement. In this study, we developed a graph neural network (GNN) framework that models electrode arrays as graphs each contact becomes a node, and functional connectivity between contacts defines the edges. We drew on two publicly available datasets from the Hospital of the University of Pennsylvania, comprising 81 patients and 212 recordings, 126 of which carried expert-confirmed SOZ annotations. Our pipeline employs GraphSAGE with self-supervised pretraining and online data augmentation. The base model achieved an area under the receiver operating characteristic curve (AUC) of 0.768, which improved to 0.785 ± 0.013 when Random Forest predictions were incorporated as an additional node feature. Most notably, the model identified 63% of true SOZ electrodes, compared with 44% for Random Forest alone even though RF produced a higher overall AUC (0.847). From a clinical standpoint, this sensitivity advantage matters: failing to flag an SOZ electrode during presurgical evaluation carries far greater consequences than including a few extra contacts for expert review. These findings suggest that GNN-based approaches, particularly hybrid designs that combine classical machine learning with graph-structured deep learning, hold genuine promise as decision-support tools in epilepsy surgical planning.

Keywords: *seizure onset zone, graph neural networks, intracranial EEG, epilepsy surgery, deep learning*

1. Introduction

Epilepsy affects an estimated 50 million individuals worldwide, and for roughly one in three, antiseizure medications fail to provide adequate seizure control (Kwan & Brodie, 2000). For these patients, surgical resection of the epileptogenic zone offers the most reliable path toward seizure freedom, with reported success rates ranging from 50% to 80% depending on the underlying etiology (Engel et al., 2012). The critical prerequisite for a favorable surgical outcome, however, is precise identification of where seizures originate. If the seizure onset zone is incompletely delineated, the resection is likely to fail.

Intracranial EEG whether acquired through stereotactically placed depth electrodes (SEEG) or subdural grid arrays (ECoG) provides direct recordings from cortical and subcortical tissue with excellent temporal and spatial fidelity. Interpreting these recordings remains a largely manual endeavor: epileptologists devote hours to examining multi-channel traces for the electrographic hallmarks of seizure onset. The process is inherently subjective, and studies have documented meaningful disagreement among experienced clinicians asked to identify the same SOZ from the same data (Varatharajah et al., 2018). This variability underscores the need for automated, data-driven methods that can complement expert judgment.

Deep learning has gained traction in neurophysiological signal analysis over the past several years, yet standard convolutional and recurrent architectures are not ideally suited to electrode arrays. The spatial layout of intracranial contacts varies from patient to patient dictated by clinical hypotheses about seizure origin and does not conform to a regular grid. Graph neural networks (GNNs) offer a natural solution to this structural irregularity: they represent electrodes as nodes in a graph and encode the functional relationships between recording sites as weighted edges (Bessadok et al., 2022). This formulation respects the topology of the data without imposing arbitrary spatial assumptions.

In this paper, we present a GNN-based pipeline for SOZ localization, developed and evaluated on data from 81 patients drawn from two publicly available iEEG repositories. The work makes five contributions: a preprocessing pipeline capable of handling multiple recording formats; a self-supervised pretraining strategy that leverages unlabeled recordings; a systematic evaluation of augmentation techniques; a head-to-head comparison of GNN architectures; and an investigation of hybrid methods that integrate classical machine learning with graph-based deep learning.

2. Materials and Methods

2.1 Datasets

We used two openly available datasets from the Hospital of the University of Pennsylvania (HUP), both hosted on OpenNeuro. The first (ds003029) includes recordings from 23 patients with drug-resistant epilepsy who underwent comprehensive presurgical evaluation. Data files are stored in BrainVision format, and SOZ annotations appear as event markers within the BIDS-standard events.tsv files. The second dataset (ds004100) is substantially larger, encompassing 58 patients who received either SEEG or ECoG monitoring prior to surgical or laser ablation treatment. Recordings are stored in European Data Format (EDF), with SOZ labels embedded in the channels.tsv metadata. Engel outcome scores are available for all surgical cases in this cohort.

In total, the combined dataset comprises 212 recordings from 81 patients, of which 126 carry expert-annotated SOZ labels (Table 1). The remaining recordings, lacking SOZ annotations, were reserved for self-supervised pretraining.

Table 1. Dataset characteristics.

Characteristic	**ds003029**	**ds004100**	**Combined**
Patients	23	58	81
Recordings	103	109	212
Recordings with SOZ labels	30	96	126
Electrode type	SEEG/ECoG	SEEG/ECoG	Mixed
Sampling rate	500–2000 Hz	500 Hz	Variable

2.2 Preprocessing Pipeline

The preprocessing pipeline was designed to accommodate both BrainVision and EDF input formats and consists of four stages. First, we extracted 60-second epochs centered on each annotated seizure onset (30 seconds before, 30 seconds after), discarding any channels flagged as bad in the recording metadata. Second, we applied notch filtering to suppress power-line interference at 60 Hz and its harmonics through the fourth (120, 180, 240 Hz); the fourth harmonic was included because 240 Hz falls near the upper edge of our subsequent bandpass and could otherwise contaminate the passband. Third, we performed common average referencing (CAR), subtracting the mean voltage across all retained electrodes at each time point to attenuate volume conduction artifacts and common-mode noise. Fourth, we applied a bandpass filter with corners at 1 Hz and 250 Hz using an elliptic design, preserving both slow cortical rhythms and the high-frequency oscillations (HFOs) that have been linked to epileptogenic tissue (Jacobs et al., 2012).

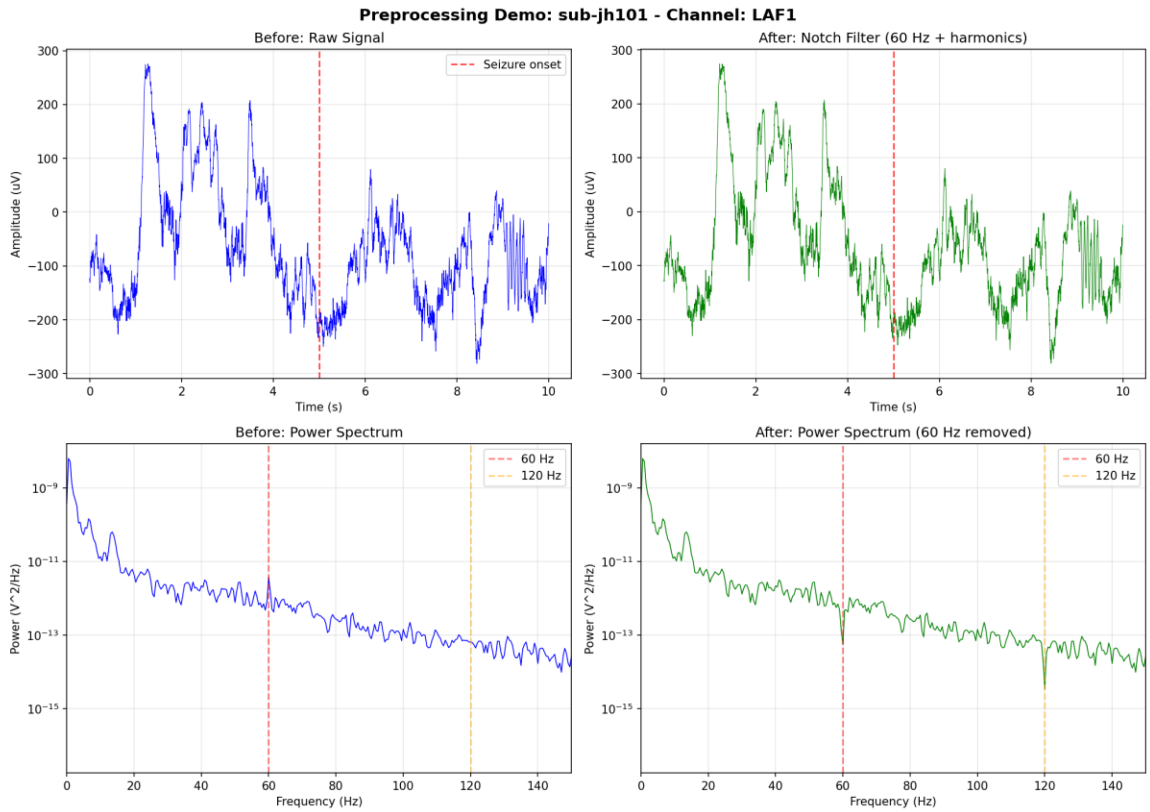

Figure 1. *Preprocessing stages illustrated on a representative iEEG trace. Top row: (left) raw signal with seizure onset marked by dashed red line, (right) after notch filtering at 60 Hz and harmonics. Bottom row: corresponding power spectra before and after notch filtering, showing attenuation of 60 Hz and 120 Hz artifacts.*

2.3 Feature Extraction

Each preprocessed recording was segmented into non-overlapping 12-second windows, from which we computed a set of features for every electrode. Spectral features comprised power in six frequency bands: delta (1-4 Hz), theta (4-8 Hz), alpha (8-12 Hz), beta (15-25 Hz), low gamma (35–50 Hz), and the HFO band (80–150 Hz), all estimated via Welch's method. Statistical features included variance, line length (the sum of absolute differences between consecutive samples), kurtosis, and skewness. We also computed relative HFO power the ratio of HFO-band power to total signal power as an additional feature, motivated by the observation that elevated HFO activity is frequently associated with epileptogenic cortex.

Features were aggregated across windows using the mean, standard deviation, and maximum, yielding 33 features per electrode. To capture inter-electrode functional connectivity, we computed Pearson correlation coefficients between all electrode pairs within each window and aggregated these using the same three summary statistics.

2.4 Graph Construction

Each recording was represented as a graph in which nodes correspond to electrodes and edges encode functional connectivity. An edge was placed between any pair of electrodes whose correlation exceeded a threshold of 0.32, a value determined through hyperparameter optimization. Node features consisted of the 33 aggregated descriptors described above, and edge weights reflected the magnitude of the pairwise connectivity. Electrode-level labels were binary: contacts identified as SOZ by clinical experts received a positive label, and all others were labeled negative. The resulting class distribution was markedly imbalanced, with SOZ contacts accounting for approximately 8.9% of all electrodes across the combined dataset.

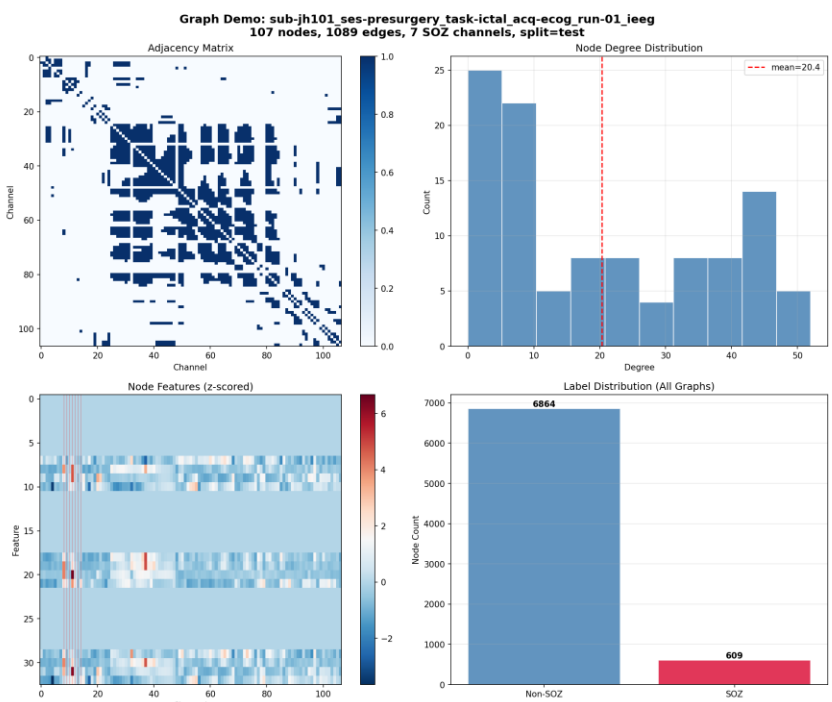

Figure 2. *Representative graph derived from an intracranial electrode array (sub-jh101, 107 nodes, 1089 edges, 7 SOZ channels). Top left: adjacency matrix reflecting the correlation-based connectivity structure. Top right: node degree distribution with mean degree of 20.4. Bottom left: z-scored node feature matrix (33 features × 107 electrodes), with SOZ contacts highlighted in red. Bottom right: label distribution across all graphs, illustrating the class imbalance (6864 non-SOZ vs. 609 SOZ nodes).*

2.5 Model Architecture

We adopted GraphSAGE (Hamilton et al., 2017) as the primary architecture. The encoder comprises two GraphSAGE convolutional layers with batch normalization, 128 hidden dimensions, and 21% dropout. Each layer updates a node's representation by sampling and aggregating features from its local neighborhood, enabling the model to learn representations that reflect both local electrographic properties and broader network context. A classification head consisting of a two-layer MLP with 64 hidden units, 55% dropout, and a sigmoid activation produces the final binary SOZ prediction for each node.

2.6 Training Strategy

Training proceeded in two phases. In the first, we pretrained the graph encoder on a self-supervised forecasting task: given features at time window t, the model learned to predict features at window t+1. This stage used 827 window pairs sampled from all available recordings, including those without SOZ labels, allowing the encoder to develop general-purpose representations of intracranial EEG dynamics before exposure to classification labels.

In the second phase, we fine-tuned the complete model (encoder plus classification head) for SOZ classification using weighted binary cross-entropy, with class weights set inversely proportional to class frequencies to address the substantial label imbalance. Online data augmentation was applied during fine-tuning: at each training step, we randomly dropped 7.6% of graph edges, added Gaussian noise ($\sigma = 0.011$) to node features, and masked 10% of feature values. Optimization was performed with AdamW, using learning rates of 1.2×10^{-4} for pretraining and 6.1×10^{-4} for fine-tuning, a cosine annealing schedule, and early stopping triggered by validation AUC.

2.7 Experimental Setup

Data were partitioned at the patient level to prevent information leakage, with 60% allocated to training, 20% to validation, and 20% to testing. This yielded 70 training graphs, 36 validation graphs, and 20 test graphs. We evaluated three GNN architectures: GraphSAGE, Graph Attention Network (GAT; Velickovic et al., 2018), and Graph Isomorphism Network (GIN; Xu et al., 2019). Hyperparameter optimization was conducted using Optuna with 50 Bayesian search trials per architecture.

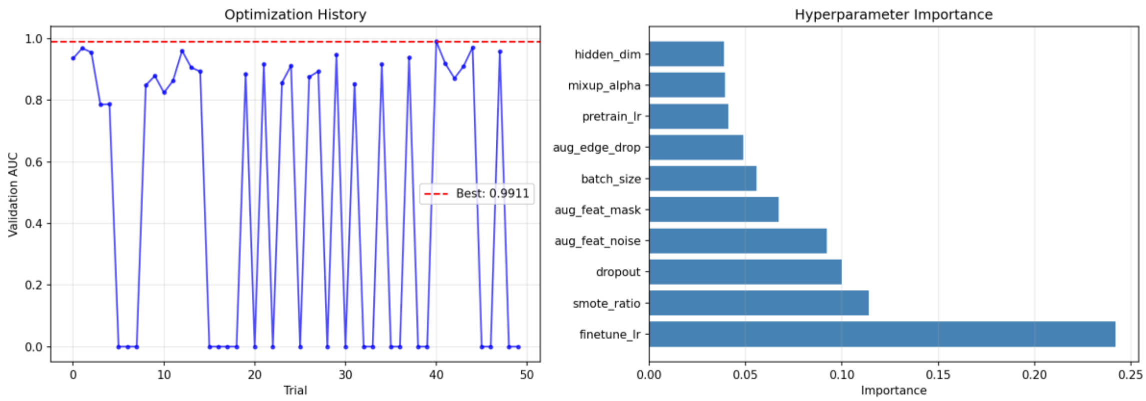

Figure 3. *Optuna hyperparameter optimization results. Left: validation AUC across 50 Bayesian search trials; best validation AUC of 0.9911 is indicated by the red dashed line. Right: ranked hyperparameter importance, showing that fine-tuning learning rate (finetune_lr) is the dominant factor, followed by SMOTE ratio and dropout rate.*

3. Results

3.1 Comparison with Classical Machine Learning

To contextualize the GNN results, we trained several classical machine learning models on the same per-electrode feature set, without any graph structure. Table 2 summarizes the performance of each approach on the held-out test set.

Table 2. Performance comparison with classical ML baselines (combined dataset).

Method	Test AUC	Test F1	Test Precision	Test Recall
Random Forest	0.847	0.373	0.326	0.436
Logistic Regression	0.741	0.336	0.238	0.571
Lasso (L1)	0.738	0.338	0.240	0.571
Ridge (L2)	0.741	0.336	0.238	0.571
LR (tuned, CV)	0.742	0.340	0.242	0.571
SVM (RBF)	0.713	0.097	0.200	0.064
GraphSAGE (ours)	0.763 ± 0.008	0.273 ± 0.007	0.175 ± 0.006	0.623 ± 0.014

A noteworthy pattern emerges from these results. Random Forest achieved the highest overall AUC (0.847), surpassing the GNN by a substantial margin. However, the GNN detected 62.3% of true SOZ electrodes, compared with 43.6% for Random Forest a relative improvement of approximately 43%. In clinical scenarios where the cost of missing a true SOZ contact is high, this sensitivity advantage carries considerable weight.

3.2 GNN Architecture Comparison

We initially benchmarked three GNN architectures on the smaller ds003029 dataset alone (Table 3). After hyperparameter tuning, GraphSAGE achieved the strongest test AUC of 0.730, outperforming GAT (0.702) and GIN (0.562). GIN notably failed to produce meaningful predictions on this dataset, yielding a test F1 of zero.

Table 3. GNN architecture comparison (ds003029 only).

Architecture	Val AUC	Test AUC	Test F1
GraphSAGE (baseline)	0.920	0.697	0.478
GraphSAGE (tuned)	0.983	0.730	0.404
GAT (tuned)	0.925	0.702	0.411
GIN	0.660	0.562	0.000

3.3 Data Augmentation Analysis

Not all augmentation strategies proved beneficial (Table 4). Online augmentation comprising edge dropout, feature noise injection, and feature masking improved the test AUC from 0.730 to 0.761, a meaningful 4.2% gain. By contrast, time-shift augmentation slightly degraded performance, mixup had a negligible effect, and SMOTE was actively detrimental, reducing AUC by 4.7%. These results suggest that augmentation strategies operating directly on the graph structure and features are better suited to this problem than techniques borrowed from other domains.

Table 4. Effect of data augmentation techniques on test AUC.

Technique	Test AUC	Change
None (baseline)	0.730	
Online augmentation	0.761	+4.2%
Time-shift	0.718	−1.6%
Mixup	0.731	+0.1%
SMOTE	0.696	−4.7%

3.4 Combined Dataset Performance

Combining both datasets provided substantially more training data 70 labeled graphs compared with approximately 20 when training on ds003029 alone and this translated into improved performance across all metrics. A single training run achieved 0.768 AUC with 65% recall. Averaged over five random seeds, the model produced an AUC of 0.763 ± 0.008 and recall of 62.3% ± 1.4% (Tables 5 and 5a).

Table 5. Performance on the combined dataset.

Metric	Single Run	5 Seeds (mean ± std)
Validation AUC	0.751	
Test AUC	0.768	0.763 ± 0.008
Test F1	0.296	0.273 ± 0.007
Test Precision	0.192	0.175 ± 0.006
Test Recall	0.650	0.623 ± 0.014

Table 5a. Per-seed results on the combined dataset.

Seed	Test AUC	Test F1	Test Recall
42	0.753	0.260	0.621
123	0.776	0.278	0.643
456	0.762	0.281	0.629
789	0.756	0.270	0.600
2024	0.768	0.274	0.621

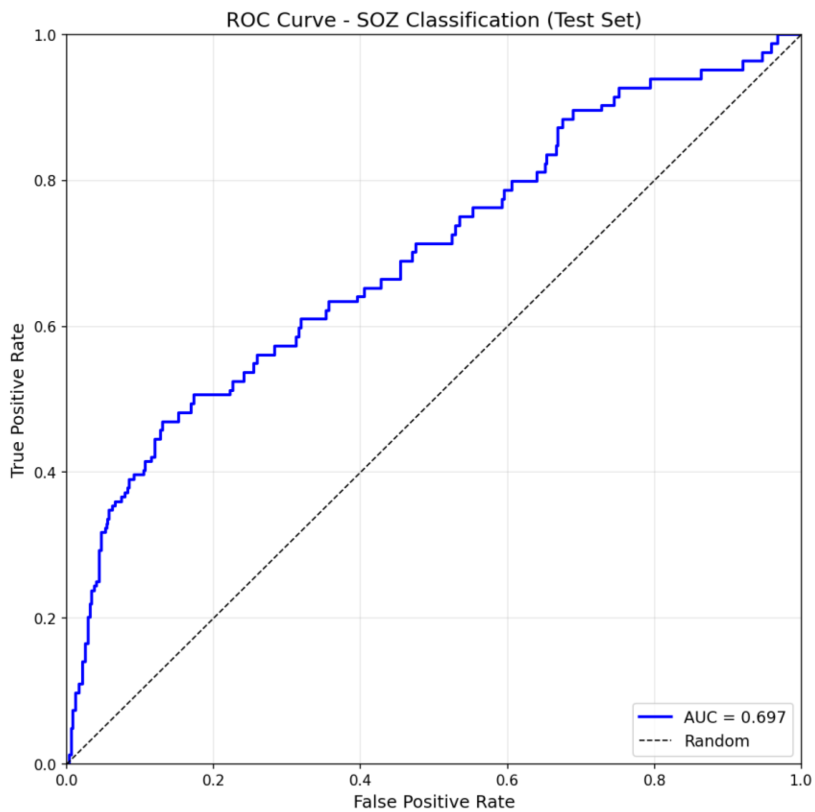

Figure 4. *Receiver operating characteristic (ROC) curve for the test set. The GraphSAGE model achieved an AUC of 0.697 on this initial single-dataset run; subsequent combined-dataset training improved this to 0.768 (see Table 5). The dashed diagonal represents chance-level discrimination.*

3.5 Experimental Progression

Table 6 traces the trajectory of model development. Each methodological addition hyperparameter tuning, augmentation, dataset expansion, and hybrid integration contributed incremental improvements, though their magnitudes varied. The single largest gains came from expanding the training set and from incorporating Random Forest predictions as a node-level feature.

Table 6. Summary of experimental progression.

Configuration	**Training Graphs**	**Test AUC**	**Test Recall**
ds003029 baseline	~20	0.697	0.47
ds003029 tuned	~20	0.730	0.40
ds003029 + augmentation	~20	0.761	0.52
Combined (single run)	70	0.768	0.650
Combined (5 seeds)	70	0.763 ± 0.008	0.623 ± 0.014
Combined + RF feature	70	0.785 ± 0.013	0.630 ± 0.052

3.6 Hybrid Approaches

The persistent AUC gap between Random Forest and the GNN motivated a series of hybrid experiments aimed at capturing the strengths of both approaches.

To establish that graph structure contributes meaningfully, we trained a multilayer perceptron (MLP) with an identical architecture but without message passing effectively treating each electrode as independent. This ablation yielded an AUC of 0.742 ± 0.009, roughly 2.8% below the GNN's 0.763, confirming that neighborhood aggregation provides a non-trivial benefit.

We next explored graph sparsity by raising the correlation threshold to 0.4, 0.5, and 0.6. All three settings produced lower AUC values (0.72–0.75), indicating that the denser connectivity afforded by the 0.32 threshold better serves the model.

A stacking ensemble combining RF and GNN predictions through a learned meta-classifier recovered much of RF's AUC advantage (0.837), but at a steep cost: recall dropped to 40%. The meta-learner effectively learned to defer to RF's more conservative probability estimates.

The most effective hybrid strategy proved to be the simplest. We trained Random Forest on the node features, then appended its predicted SOZ probability as an additional input feature when training the GNN. This approach achieved an AUC of 0.785 ± 0.013 with recall of 63% ± 5% capturing most of the AUC improvement while preserving the sensitivity advantage that makes the GNN clinically useful (Table 7).

Table 7. Comparison of hybrid approaches.

Method	Test AUC	Test F1	Test Precision	Test Recall
Random Forest	0.847	0.373	0.326	0.436
GNN (baseline)	0.763 ± 0.008	0.273 ± 0.007	0.175 ± 0.006	0.623 ± 0.014
MLP (no graph)	0.742 ± 0.009	0.286 ± 0.007	0.192 ± 0.007	0.566 ± 0.015
Stacking (RF + GNN)	0.837	0.336	0.290	0.400
Stacking (full)	0.829	0.318	0.292	0.350
GNN + RF feature	0.785 ± 0.013	0.293 ± 0.012	0.192 ± 0.015	0.630 ± 0.052

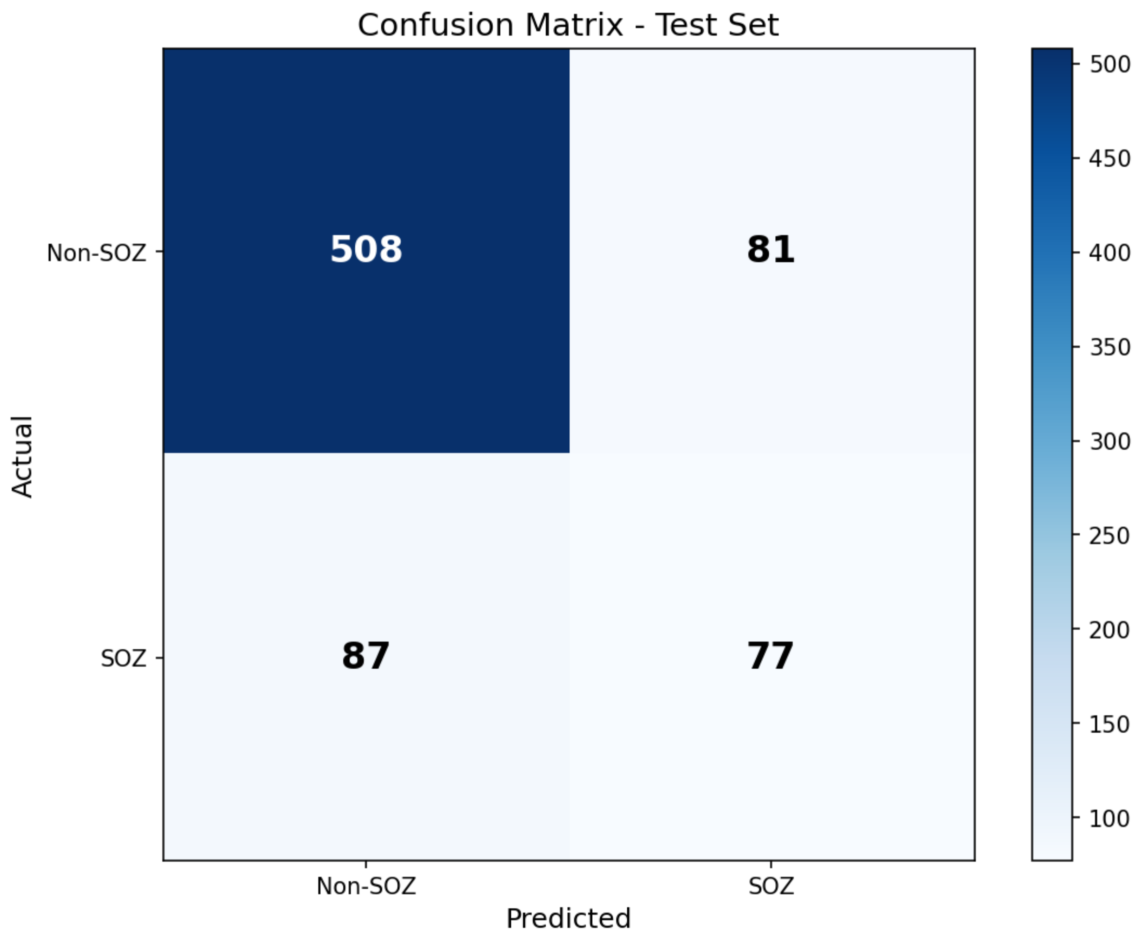

Figure 5. *Confusion matrix for the test set. Of 164 true SOZ electrodes, 77 were correctly identified (true positives) and 87 were missed (false negatives). Of 589 non-SOZ electrodes, 508 were correctly classified and 81 were falsely flagged as SOZ. These counts correspond to a recall of 47% and precision of 49% for this specific run.*

3.7 Extension to Surgical Outcome Prediction

While electrode-level SOZ localization addresses the question of where seizures originate, an equally pressing clinical question is whether a given patient will become seizure-free following resective surgery. We therefore extended our framework to predict patient-level surgical outcomes, defined as Engel Class I (seizure-free) versus Engel Class II–IV (not seizure-free).

3.7.1 Patient Cohort and Label Distribution

From the combined dataset, 57 patients had documented Engel outcome scores following surgery. Of these, 40 achieved seizure freedom (Engel I) and 17 did not (Engel II–IV). This represents a moderately imbalanced classification problem with a 70:30 class ratio. Many patients contributed multiple iEEG recordings—captured during different seizure events or monitoring sessions—yielding a total of 127 recording-level graphs linked to these 57 patients.

3.7.2 Multi-Instance Learning Formulation

A fundamental challenge in patient-level prediction is that outcome labels exist at the patient level, while our graph representations exist at the recording level. Assigning the patient's outcome label to each individual recording would conflate the unit of analysis, risk data leakage across recordings from the same patient, and ignore the fact that different recordings may carry varying amounts of prognostic information.

We addressed this through a multi-instance learning (MIL) formulation. Each patient is treated as a "bag" containing one or more recording graphs. A shared graph encoder processes each recording independently, producing a fixed-dimensional embedding. These embeddings are then aggregated across recordings using an attention mechanism that learns to weight recordings according to their relevance for outcome prediction. The aggregated patient-level representation is passed through a classification head to produce a single seizure-free probability per patient.

3.7.3 Model Configuration

The recording encoder consisted of a single GraphSAGE layer with 32 hidden dimensions—deliberately smaller than the SOZ localization model to reduce overfitting given the limited patient count. The attention module comprised a two-layer MLP projecting each recording embedding to a scalar attention logit. We applied 40% dropout throughout and trained with AdamW using a learning rate of 1×10^{-3} and weight decay of 5×10^{-4}.

Notably, incorporating clinical features (lesion status, prior therapy, surgical target) degraded performance rather than improving it. This counterintuitive result stemmed from substantial missing data in the ds003029 subset: imputation introduced noise that outweighed any informational benefit. All reported results therefore use iEEG-derived features exclusively.

3.7.4 Evaluation Strategy

Given the small sample size, a single train/validation/test split would produce unstable estimates highly dependent on which patients happened to fall into each partition. We therefore adopted a repeated stratified cross-validation design: 5-fold cross-validation repeated 5 times with different random seeds, yielding 25 total evaluations. Stratification ensured that each fold preserved the approximate 70:30 class ratio. No patient appeared in both training and test sets within any fold, eliminating data leakage.

3.7.5 Results

The MIL-GNN model achieved an AUC of 0.807 ± 0.121, substantially outperforming Random Forest trained on aggregated recording features (AUC 0.541). The high standard deviation reflects the inherent variability of small-sample evaluation—with only 11–12 patients per fold, individual fold results ranged from 0.65 to 0.95. Table 8 summarizes the patient-level outcome prediction results.

Table 8. Patient-level surgical outcome prediction (5-fold × 5 repeats).

Model	AUC	Accuracy	F1	Recall
Random Forest	0.541 ± 0.12	0.57 ± 0.11	0.73 ± 0.09	1.00
MIL-GNN (iEEG only)	0.807 ± 0.121	0.64 ± 0.15	0.67 ± 0.27	0.69 ± 0.32
MIL-GNN + clinical	0.761 ± 0.170	0.62 ± 0.14	0.72 ± 0.14	0.73 ± 0.22

Several alternative approaches were explored but did not improve upon the baseline MIL model. An ensemble combining MIL predictions with Random Forest achieved AUC of approximately 0.75, limited by the weak RF component. Transfer learning from the SOZ localization task—using the pretrained SOZ encoder as initialization—yielded AUC of 0.750, suggesting partial but incomplete benefit from the related task.

4. Discussion

The GNN pipeline presented here demonstrates that graph-structured deep learning can identify SOZ electrodes from intracranial EEG with clinically relevant accuracy. The AUC of 0.768 falls within the 0.65–0.85 range reported by prior machine learning studies addressing this task (Varatharajah et al., 2018; Bernabei et al., 2022), and the hybrid variant pushes this to 0.785.

4.1 Clinical Implications

The most clinically significant finding is the model's recall of 62.3%. In the context of presurgical evaluation, the asymmetry of errors is stark: an SOZ electrode that is missed may lead to an incomplete resection and continued seizures, whereas a false positive merely adds a contact to the set that clinicians review. Random Forest, despite its superior AUC, identified only 43.6% of true SOZ contacts. The GNN's 43% relative improvement in sensitivity could, if validated prospectively, translate into more complete resections and better postoperative outcomes.

We attribute this sensitivity advantage to the GNN's capacity for capturing relational information. SOZ electrodes tend to exhibit distinctive connectivity patterns with their neighbors patterns that are invisible to methods that treat each contact in isolation. By aggregating information from surrounding nodes, the GNN can recognize these network-level signatures even when the local features of an individual electrode are ambiguous.

The tradeoff is precision. At 17.5%, the model generates a substantial number of false positives. In practice, however, this is a manageable burden. A clinical workflow might use the model's output as a first-pass screen, flagging contacts for expert review rather than dictating the final surgical plan. Under this paradigm, high sensitivity is the priority, and the additional review time imposed by false positives is a reasonable cost.

4.2 Understanding the Performance Gap

That Random Forest outperformed the GNN on AUC was not what we anticipated. Several factors likely contribute to this result. First, the training set is small by deep learning standards 70 graphs with approximately 6,200 labeled nodes. Neural networks are known to struggle with limited data, tending to memorize training patterns rather than learn generalizable representations (Shorten & Khoshgoftaar, 2019). Random Forest, by contrast, benefits from the regularizing effects of bagging and feature subsampling, which naturally mitigate overfitting in low-data regimes (Katzmann et al., 2020).

Second, the engineered features are highly informative. Band powers, HFO activity, line length, and statistical moments encode decades of accumulated clinical knowledge about epileptogenic tissue. When the input features already carry strong discriminative signal, the additional representational capacity afforded by graph convolutions may yield diminishing returns (Chen et al., 2020).

Third, graph construction remains an open methodological question. Our correlation-based thresholding approach, while straightforward, may not capture the connectivity patterns most relevant to SOZ identification. If the graph topology is noisy or incomplete, message passing can propagate misleading information through the network, undermining rather than enhancing node representations.

Despite these considerations, the GNN's sensitivity advantage is robust and reproducible across random seeds. The two approaches appear to operate in complementary regimes: Random Forest achieves strong overall discrimination by being selective, while the GNN casts a wider net that captures true SOZ contacts at the expense of specificity.

4.3 The Value of Multi-Site Data

Combining two datasets even ones from the same institution collected under different protocols produced a larger performance gain than any augmentation technique we evaluated. This observation reinforces a broader lesson in clinical machine learning: data diversity matters at least as much as data volume. Initiatives such as OpenNeuro that facilitate data sharing stand to accelerate progress in this field considerably.

4.4 Clinical Relevance of Outcome Prediction

The extension to patient-level outcome prediction addresses a question that weighs heavily on both clinicians and patients during presurgical counseling: what are the chances that this surgery will work? An AUC of 0.807, while imperfect, suggests that iEEG-derived graph features carry prognostic information beyond what is captured by clinical variables alone. If validated in larger

cohorts, such a model could inform shared decision-making by providing a data-driven probability estimate to complement clinical judgment.
The attention mechanism embedded in the MIL architecture offers an additional benefit: interpretability. By examining which recordings receive the highest attention weights for a given patient, clinicians might identify the seizures or monitoring epochs that are most informative for prognosis.

4.5 Limitations of Outcome Prediction

The outcome prediction results should be interpreted with appropriate caution. The sample of 57 patients, while sufficient to demonstrate feasibility, is too small for reliable generalization. The wide confidence interval (±0.121) reflects genuine uncertainty about the model's true performance. Furthermore, the retrospective design cannot account for selection effects: patients who underwent surgery represent a subset deemed to be good surgical candidates, and results may not extrapolate to a broader population.

4.6 Limitations and Future Directions

Several limitations should be acknowledged. All data originated from a single institution, and cross-site generalizability has not been tested. The overall sample size, while sufficient for preliminary evaluation, remains modest for deep learning. Performance was assessed at the electrode level rather than the patient level, which may overstate clinical utility. Additionally, our graph construction relied on a single connectivity measure (Pearson correlation) with a fixed threshold; alternative measures or adaptive thresholding schemes warrant investigation.

Future work should prioritize multi-center validation, integration of anatomical electrode coordinates to complement functional connectivity, and extension of the framework to seizure prediction tasks. The application of GNN explainability techniques could also illuminate which features and network motifs are most informative for SOZ identification, potentially yielding insights that inform clinical interpretation.

5. Conclusion

We developed a graph neural network framework for automated seizure onset zone localization in patients undergoing intracranial EEG monitoring for drug-resistant epilepsy. Using two publicly available datasets encompassing 81 patients and 126 labeled recordings, combined with self-supervised pretraining and online augmentation, the GraphSAGE model achieved an AUC of 0.768 with 62–65% recall. While Random Forest produced a higher AUC (0.847), the GNN identified substantially more true SOZ electrodes (62.3% versus 43.6%), reflecting a sensitivity advantage that is directly relevant to surgical planning.

Three findings from the hybrid experiments merit emphasis. Graph structure contributes a measurable 2.8% AUC improvement over an equivalent non-graph model. Stacking Random Forest and GNN predictions improves AUC but sacrifices recall. Incorporating RF's predictions as a node-level feature achieves the best of both: an AUC of 0.785 ± 0.013 with recall of 63% ± 5%.

Taken together, these results indicate that graph-based deep learning particularly when integrated with classical machine learning holds meaningful promise as a clinical decision-support tool for epilepsy surgery. Prospective validation, cross-institutional testing, and integration into clinical workflows represent the logical next steps toward translating these findings into practice.

Beyond electrode-level SOZ localization, we demonstrated that the same graph-based framework can be extended to predict patient-level surgical outcomes. Using a multi-instance learning formulation that aggregates information across multiple recordings per patient, the model achieved an AUC of 0.807 ± 0.121 for distinguishing seizure-free from non-seizure-free outcomes. While the sample size precludes definitive conclusions, these preliminary results suggest that graph neural networks can extract prognostically relevant patterns from intracranial EEG—patterns that may ultimately inform surgical decision-making if validated in larger, multi-center cohorts.

Data and Code Availability

Both datasets are publicly available on OpenNeuro:

ds003029 (https://openneuro.org/datasets/ds003029)

ds004100 (https://openneuro.org/datasets/ds004100).

Code is available at: https://github.com/abtinunmc/GNN-PROJECT

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Competing Interests

The author declares no competing interests.

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Ethics Statement

This study is based exclusively on de-identified, publicly available intracranial EEG datasets (OpenNeuro ds003029 and ds004100). No new patient data were collected, and no direct patient contact occurred. Accordingly, this work did not require independent institutional review board (IRB) approval. The original datasets were collected under appropriate ethical oversight at their respective institutions, as described in the associated publications.